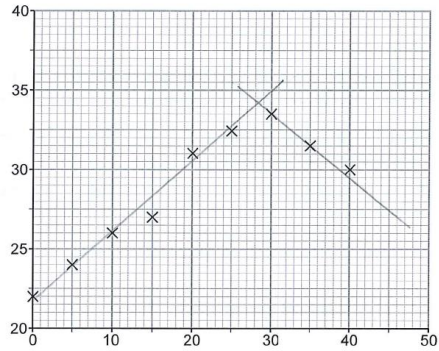


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		pipette / burette	1			1		1
		(ii)		more H ⁺ / methanoic acid is neutralised			1	1		1
		(iii)		all the H ⁺ / methanoic acid has been neutralised (1) award (1) for either of following excess OH ⁻ added cools the solution same amount of energy has to heat a greater volume neutral answer – cools to room temperature			2	2		2
		(iv)		two straight lines drawn through points and intersecting			1	1	1	1
		(v)		award (1) for volume and (1) for temperature increase within the following ranges volume ⇨ 27.0-30.0 cm ³ temperature increase ⇨ 11.0-13.0 °C 		2		2	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(vi)		based on graph in part (iv) total volume of solution heated = $25.0 + 28.0 = 53.0$ (1) heat released = $53.0 \times 4.18 \times 12.0 = 2660$ J (1) ecf possible when 25.0 or 28.0 used as the mass or when the maximum temperature is used instead of the temperature <u>increase</u>		2		2	1	
		(vii)		moles methanoic acid = $\frac{24.7}{46.02} = 0.537$ (1) moles in $25.0 \text{ cm}^3 = 0.0537$ (1) enthalpy change = $-\frac{2660}{0.0537 \times 1000} = -49.5 \text{ kJ mol}^{-1}$ (1) ecf possible when an attempt has been made to calculate a number of moles		3		3	2	
	(b)			more H^+ is available / there is a higher concentration of H^+ (1) award (1) for either of following hydrochloric acid is strong and methanoic acid is weak hydrochloric acid is stronger than methanoic acid neutral answers – more acidic / donates H^+ more easily	1		1	2		
	(c)	(i)		$\text{CuCO}_3(\text{s}) + 2\text{HCOOH}(\text{aq}) \rightarrow (\text{HCOO})_2\text{Cu}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ formulae of reactants and products (1) state symbols and balancing (1)		2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		award (1) each for any two of following effervescence / bubbles / fizzing blue solution forms solid disappears neutral answer – colour changes	2			2		2
				Question 10 total	4	9	5	18	5	7